



# Introduction to Statistics

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# What to Expect

- Why should you care about statistics?
- Basic overview of Frequentist and Bayesian approaches
- Relevant history
- Practical implementation (MCMC)
- Interpretation of results

# Statistics: why should you care?

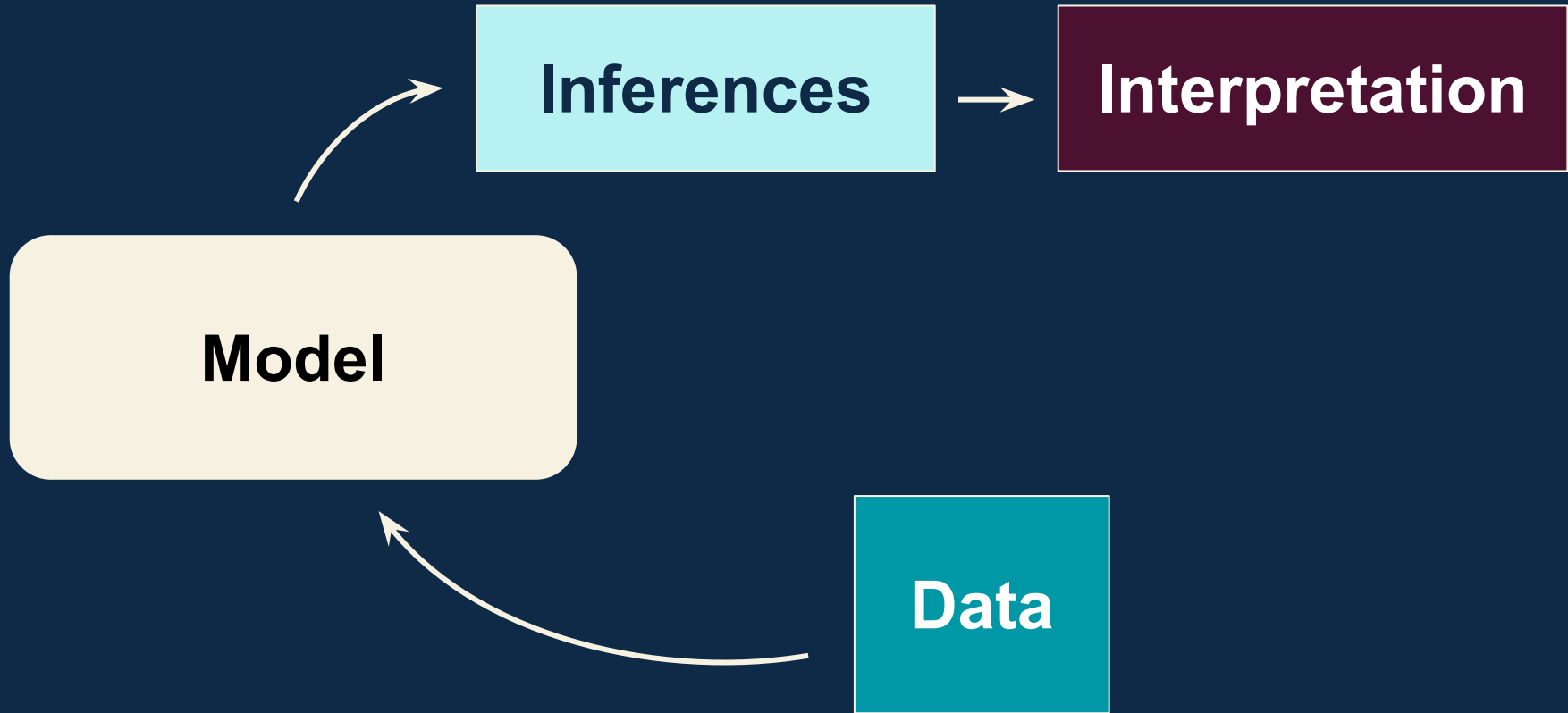
# Statistics: why should you care?

have data

want answers

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have data      statistics! →      want answers



# Two Schools of Thought

**Frequentist**

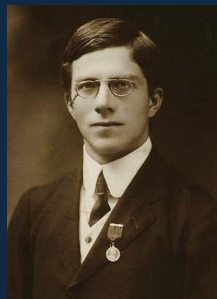
**Bayesian**

# Two Schools of Thought

## Frequentist

- Model is fixed
- Data is probabilistic
- “Given my model of the universe, what is the probability I observe a particular random realization of the data?”

## Bayesian



Sir Ronald Fisher (1890-1962)

(wikipedia commons)

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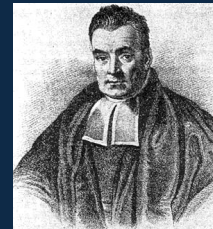
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## Bayesian

- Data is fixed
- Model is probabilistic
- “Given my observed data, what is the probability that my model parameters take on a particular set of values?”
- **Prior** information is incorporated



Rev. Thomas Bayes (1701-1761)  
(wikipedia commons)

# Bayes' Theorem

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**Joint Probability:**  $p(A,B)$

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yielding **Bayes' Theorem:**  $p(A|B) = \frac{p(B|A) p(A)}{p(B)}$

# Bayes' Theorem

**connects:**

- the **posterior**       $p(\text{model} \mid \text{data})$
- the **likelihood**     $p(\text{data} \mid \text{model})$
- the **prior**           $p(\text{model})$
- the **evidence**       $p(\text{data})$

$$p(\text{model} \mid \text{data}) = \frac{p(\text{data} \mid \text{model}) p(\text{model})}{p(\text{data})}$$

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$$p(\text{model} \mid \text{data}) \propto p(\text{data} \mid \text{model}) p(\text{model})$$

## Sidebar: The Evidence

$p(\text{data})$

- can be **difficult to compute**
- doesn't depend on our model...
- is often relegated to a **proportionality**
- but is useful for **model comparison!**

# Example: A Fair Coin

Question: we have flipped a coin twice, and both times it landed on heads. What is the probability the coin is fair?



# Example: A Fair Coin

## Frequentist Approach

Model: a fair coin

$$p(H) = 0.5$$

$$p(T) = 1 - p(H) = 0.5$$

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Probability of  $N=2$  heads given a fair coin is then

$$p(2xH) = p(H) \times p(H|H) = p(H)^2 = 0.25$$

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Model: infer whether the coin is fair

**prior**  $p(\text{fair}) = 0.99$  (most coins are fair!)

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## Bayesian Approach

**evidence** is  $p(2xH)$ :

$$\begin{aligned} p(2xH) &= \int p(2xH|\text{coin}) p(\text{coin}) d\text{coin} \\ &= p(2xH|\text{fair}) p(\text{fair}) + p(2xH|\text{biased}) p(\text{biased}) \end{aligned}$$

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**simplest model:** biased coin always flips heads  $\rightarrow p(2xH|\text{biased}) = 1$

$$p(2xH) = 0.25 * 0.99 + 1 * 0.01 = 0.2575$$

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**posterior** is then

$$\begin{aligned} p(\text{fair}|2xH) &= p(2xH|\text{fair})p(\text{fair})/p(2xH) \\ &= 0.25 * 0.99 / 0.2575 \\ &= \mathbf{0.96} \end{aligned}$$

# Try it out! In small groups:

|               | You Have the Disease | You don't have the disease |
|---------------|----------------------|----------------------------|
| Positive test | True Positive        | False Positive             |
| Negative test | False Negative       | True Negative              |



A rare disease (“RD”) has a sensitive test with a false positive rate of 1% and a false negative rate of 1%. A patient tests positive for the disease.

## Exercise 1

Compute the frequentist probability that the patient has the disease given a positive test.

How does this change if the patient is tested twice and:

- both are positive?
- one is positive and one is negative?

## Exercise 2

RD has an occurrence rate in the population of 1 per 10,000 people.

Incorporating this information, compute the Bayesian probability that the patient has the disease given a positive test.

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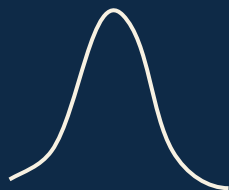
Need a **computational** approach!

- Blunt for grid search can work for a small parameter space
- Scales as  $O(n^{N_{\text{dim}}})$  (aka horribly)



**MCMC**

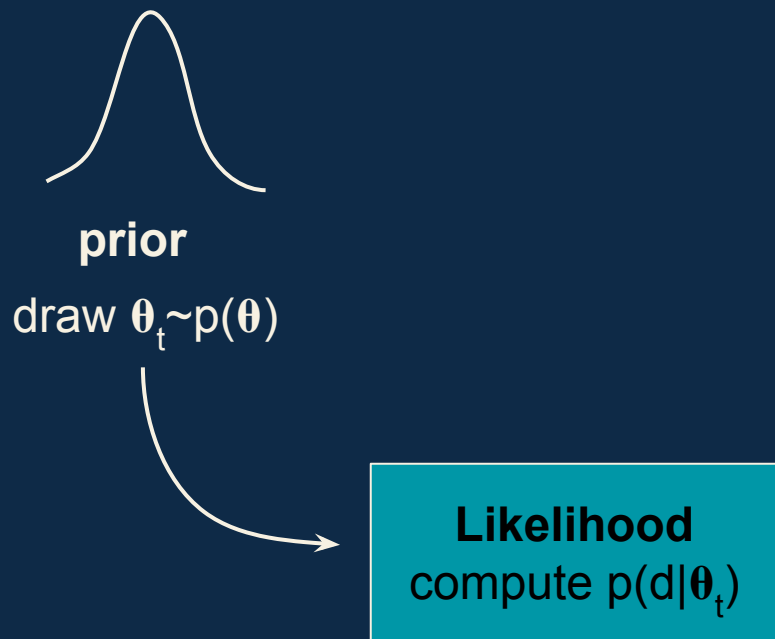
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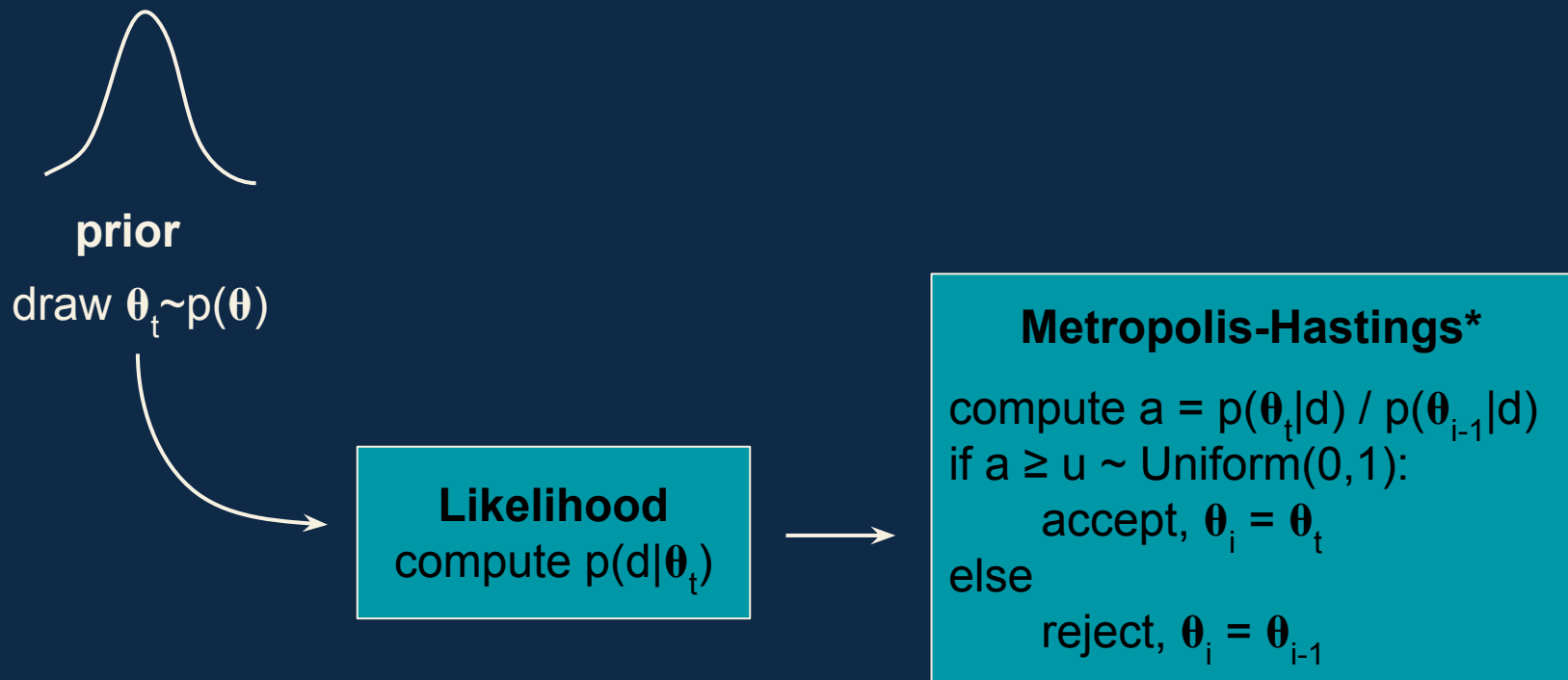
**prior**

draw  $\theta_t \sim p(\theta)$

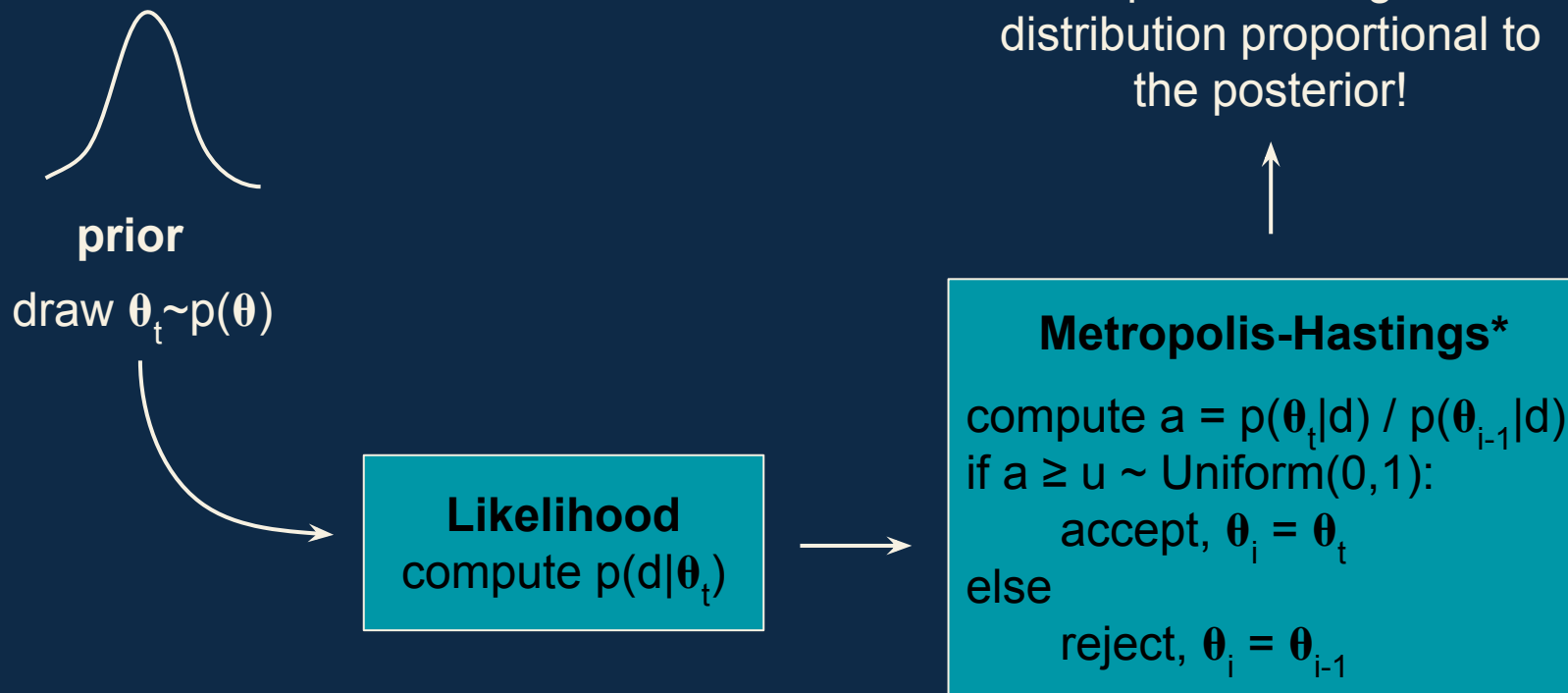
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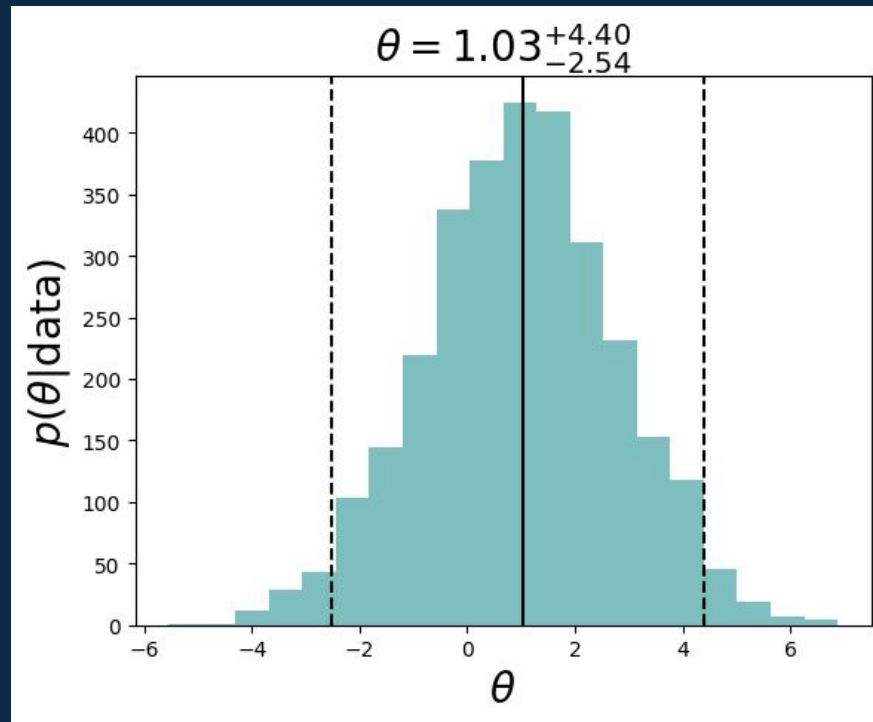
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# Interpreting a posterior

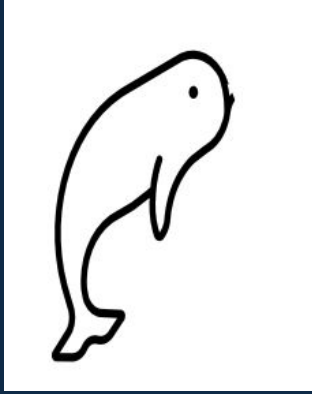
X% credible interval (C.I.):

- X% probability that the true value falls within the interval
- Common choices: 90%, 95%
- “68–95–99.7” rule corresponds to 1, 2, and 3-sigma intervals in 1D
- In 2+ dimensions: “credible region”

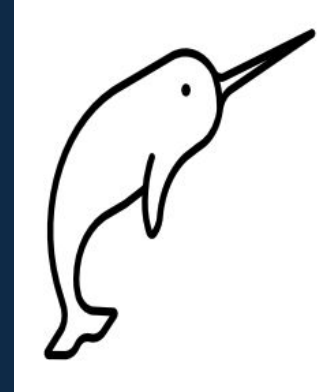


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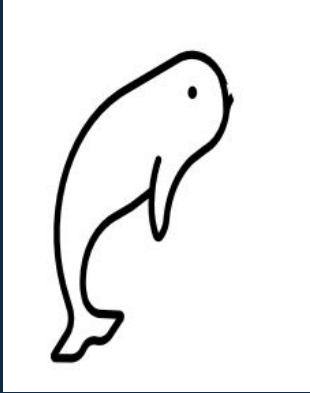


Model 1

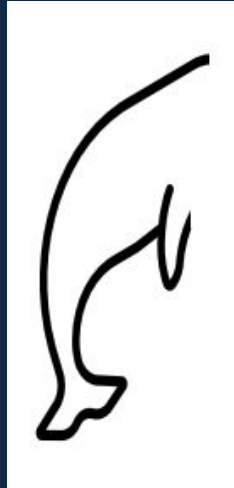


Model 2

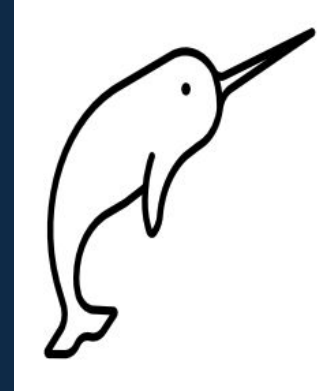
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Model 1



data.



Model 2

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**Bayes Factor**       $B = \frac{Z_1}{Z_2}$

Where  $Z_M$  is the **evidence** of model  $M$

$$Z_M = \int p(d|\theta_M) p(\theta_M) d\theta_M$$

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Bayes factors are **relative**!

High Bayes factor  $\neq$  correct model!

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$p(\text{model})$

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